

Graph World Model for Energy-Efficient Hybrid Beamforming in LEO Satellite Networks

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Abstract—Rapid channel variations in Low Earth Orbit (LEO) satellite networks require frequent updates of the digital precoder. Conventional iterative hybrid beamforming (HBF) schemes may therefore incur substantial online computational cost, whereas learning-based alternatives may require extensive environment interactions. This paper proposes a graph world model for resource allocation (GWM-RA), a model-based learning framework for energy-efficient, surrogate-assisted non-iterative digital precoding in a two-timescale HBF architecture. Specifically, analog beam steering and scheduler-provided user association are configured on a long timescale using slowly varying statistical channel information, while the short-timescale digital precoder is generated through a single forward pass conditioned on instantaneous channel state information (CSI). Under the considered independent and identically distributed (i.i.d.) block-fading setting, the GWM-RA world model is trained using environment-generated rate and power labels to approximate the immediate action–performance relation conditioned on the current graph state and a candidate digital-precoding action. This differentiable performance surrogate provides one-step estimates of the reward and constraints for policy optimization, thereby reducing the number of true-environment evaluations required under the adopted training protocol. Numerical results show that GWM-RA achieves higher energy efficiency than the considered conventional optimization algorithms and standard model-free reinforcement learning baselines. Under the same training protocol, GWM-RA reaches the target energy-efficiency threshold with over 60% fewer environment interactions than the matched GNN-MFRL counterpart.

Index Terms—Energy efficiency, heterogeneous graph neural networks, hybrid beamforming, satellite-terrestrial integrated networks, world model.

I. INTRODUCTION

SATELLITE-TERRESTRIAL integrated networks (STINs) are expected to provide wide-area connectivity and enhance network resilience in sixth-generation (6G) wireless systems [1], [2], [3]. Low Earth Orbit (LEO) constellations can provide low-latency services [4], but require large-scale

antenna arrays and hybrid beamforming (HBF) to compensate for severe free-space path loss [5]. Although HBF architectures reduce hardware costs and static power consumption, energy-efficient HBF design remains challenging in dynamic LEO environments [6].

The high orbital velocity of LEO satellites causes rapid channel variations and beam-footprint drift [4]. Effective interference management in dynamic LEO downlinks therefore requires the digital precoder to adapt to short-timescale channel state information (CSI) [7], [8]. The main computational burden arises from repeatedly solving CSI-dependent iterative optimization problems on power-constrained satellite payloads [9], [10]. Model-free learning avoids online iterative optimization after training, but policy improvement may require extensive environment interactions and incur additional adaptation overhead.

A. Related Work

Existing research on resource allocation for LEO STINs can be broadly categorized into deterministic optimization, model-free deep reinforcement learning (DRL), and model-based DRL. We discuss these paradigms in terms of three factors relevant to energy-efficient operation on power-constrained satellites: online computational cost, reliance on frequent CSI acquisition, and the ability to model non-local interference explicitly.

1) *Deterministic Optimization*: Conventional optimization approaches generally address the non-convex HBF energy efficiency problem via iterative numerical algorithms. To manage CSI uncertainty, robust HBF designs have been reformulated into convex semidefinite programming problems [11]. Addressing the coupling between analog and digital domains, joint optimization frameworks for user scheduling and resource allocation typically employ alternating minimization techniques [12]. Recent studies have extended these iterative methods to active RIS-assisted transmission [13] and Age-of-Information (AoI)-aware resource allocation [14] to improve system energy efficiency. Although schemes based on slowly varying statistical CSI can reduce dependence on instantaneous CSI [5], they may suffer performance degradation in highly dynamic LEO settings. The practical challenge is therefore to exploit instantaneous CSI without incurring excessive iterative-computation delay. Optimization-based resource management has also been applied to satellite edge com-

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puting to maximize energy efficiency under strict resource constraints [15].

2) *Model-Free Deep Reinforcement Learning*: To avoid the online computational burden of deterministic optimization, model-free DRL methods have been widely investigated [16]. DRL agents learn resource-allocation policies through direct interaction with the environment without requiring an explicit environment model. In satellite communications, risk-aware DRL frameworks have been developed for robust HBF under channel uncertainty [17]. Multi-Agent DRL (MADRL) has also been employed to address decentralized problems such as dynamic spectrum sharing in integrated networks [18] and joint optimization of beam patterns and bandwidth in beam hopping systems [19], [20]. In highly dynamic LEO networks, the varying numbers of visible satellites and scheduled users, together with their interference relationships, naturally form a heterogeneous graph. Conventional multilayer perceptrons (MLPs) are less suited to this setting because they typically require fixed-size inputs and do not explicitly represent structural dependencies under varying topologies. Graph neural networks (GNNs) have therefore been integrated into DRL frameworks [21]. Through message passing, GNNs provide a structural inductive bias for modeling non-local interference patterns [22].

Recent studies have applied GNN-based policies to power allocation in LEO networks, with meta-learning used to improve generalization across traffic distributions [23]. Beyond graph-based representation learning, model-free RL can be enhanced through exploration-oriented and architectural designs. For example, recent split-learning resource-management studies have incorporated intrinsic-curiosity rewards and cross-attention into soft actor-critic training to accelerate exploration and convergence [24]. Although these mechanisms improve model-free training, policy improvement still relies on rewards obtained through environment interaction. In satellite HBF, repeated CSI-dependent evaluation and adaptation can incur signaling and computational overhead; hence, sample efficiency remains an important design consideration. This motivates a learning paradigm that retains the adaptivity of DRL while reducing the number of true-environment evaluations required during policy improvement.

3) *Model-Based Deep Reinforcement Learning*: Model-based RL, including the world model framework, provides a means of improving the sample efficiency of model-free methods [25]. Model-based RL commonly improves policies by learning action-dependent environment models and using them for imagined policy updates. In the independent and identically distributed (i.i.d.) block-fading setting considered in this paper, the next short-slot channel realization is independent of the current digital-precoding action. Consequently, a next-state transition model would mainly describe action-independent channel variation and would not supply the policy-relevant action-dependent signal needed for the current-slot beamforming decision. In GWM-RA, the world model is therefore formulated as a task-level performance surrogate that maps the current graph-structured interference state and a candidate digital-precoding action to immediate user-wise rates and total power. The resulting differentiable surrogate supports one-step

reward and constraint evaluation for sample-efficient policy improvement, without introducing recurrent latent dynamics or multi-step rollouts.

The potential of model-based RL to improve sample efficiency has also been recognized in wireless resource allocation. For instance, a model-based RL framework was proposed for RAN slicing to satisfy service-level agreements while improving resource efficiency and limiting extensive online exploration [26]. Learning interactive simulators is also an active research direction for complex control tasks [27]. Recent surveys have further discussed generative AI as an enabler of DRL in future wireless networks [28]. Together, these studies motivate model-based learning for sample-efficient dynamic resource management. However, topology-aware model-based learning that explicitly exploits graph-structured interference for energy-efficient HBF in LEO satellite networks remains largely unexplored.

B. Motivation and Contributions

The preceding discussion identifies two practical requirements for LEO beamforming: low-complexity short-timescale precoding and sample-efficient policy training. Iterative algorithms impose repeated online computation on satellite payloads, whereas model-free learning may require numerous environment interactions and long adaptation times. The digital-precoding policy should therefore use instantaneous CSI without iterative online optimization while accommodating changing interference topologies within the considered topology family.

This paper proposes a graph world model for resource allocation (GWM-RA), a model-based learning framework for the short-slot digital-precoding component of energy-efficient HBF under fixed long-timescale analog precoding and scheduler-provided association. Its core module is a topology-aware performance world model that learns the immediate mapping from the graph-structured state and a candidate digital-precoding action to rate and power quantities used for surrogate-assisted, non-iterative optimization.

Table I summarizes the major notation used throughout the paper.

The primary contributions of this work are summarized as follows.

- 1) We propose a surrogate-assisted HBF method in which policy improvement is guided by a learned performance world model instead of repeated true-environment reward queries. The world model approximates the immediate action-performance mapping, yielding differentiable rate, power, reward, and constraint information for actor updates, while online digital precoding is still generated through a single policy-network forward pass.
- 2) We develop a topology-aware performance world model for multi-satellite interference environments represented as heterogeneous graphs. The GNN encoder forms an intermediate latent representation of the current graph state, and the prediction heads are supervised to map this representation together with a candidate digital-precoding action to user-wise rates and total

TABLE I
LIST OF MAJOR NOTATIONS

Symbol	Description
<i>System and Channel Parameters</i>	
$\mathcal{S}$	Set of satellites, $ \mathcal{S} = S$
$\mathcal{K}$	Set of ground users, $ \mathcal{K} = K$
$\mathcal{V}_k$	Visible-satellite set for user k
$\mathcal{K}_s$	User set served by satellite s
N_{TX}	Number of antennas at satellite s
N_{RF}	Number of RF chains at satellite s
$\mathbf{h}_{s,k}$	Channel vector ($s \rightarrow k$)
$\mathbf{a}(\theta, \phi)$	Array steering vector
σ^2	AWGN power
<i>Beamforming and Signal Parameters</i>	
$\mathcal{D}$	Codebook of analog beams
$\mathbf{F}_s^{\text{RF}}$	Analog precoder of satellite s
$\mathbf{F}_s^{\text{BB}}$	Digital baseband precoder
$\mathbf{f}_k^{\text{BB}}$	Digital precoder for user k
$\mathbf{w}_k$	Effective precoder for user k
d_k	Data symbol, $\mathbb{E}[d_k ^2] = 1$
p_k	Allocated power for user k
y_k	Received signal at user k
Γ_k	SINR of user k
R_k	Achievable rate of user k
<i>Optimization and Power Parameters</i>	
$R_{\text{min}}^{\text{min}}$	Minimum required rate per user
P_s^{tot}	Total power at satellite s
P_s^{max}	Maximum power budget of satellite s
η	Power-amplifier efficiency
P_s^{RF}	Power per RF chain
P_0^{PS}	Power per phase shifter
c_{eff}	Rate-dependent coefficient
P_c	Constant circuit power
EE	Energy efficiency
$\alpha_{s,k}$	Association variable
<i>MDP and GWM-RA Parameters</i>	
s_t	State at time slot t
a_t	Action at time slot t
r_t	Immediate reward at slot t
$\mathcal{G}_t$	Graph state representation
$\mathbf{z}_t$	Latent state vector
$\mathcal{B}$	Replay buffer
$\hat{\mathbf{R}}_t, \hat{P}_t$	Predicted user-wise rates and total power

power over the considered family of simulated LEO topologies.

- 3) We demonstrate through simulations that the proposed GWM-RA approach achieves higher energy efficiency than the considered iterative optimization benchmarks and standard model-free DRL counterparts. In particular, under the adopted training and adaptation protocol, GWM-RA requires over 60% fewer environment interactions than the matched standard GNN-MFRL counterpart to reach the energy-efficiency threshold. These results support the sample-efficiency benefit of surrogate-assisted policy improvement within the evaluated baseline set, rather than claiming dominance over all possible enhanced model-free RL designs.

C. Paper Organization

The remainder of this paper is organized as follows. Section II introduces the LEO satellite network architecture and the power consumption model, formulating the energy-efficient HBF problem. Section III presents the proposed

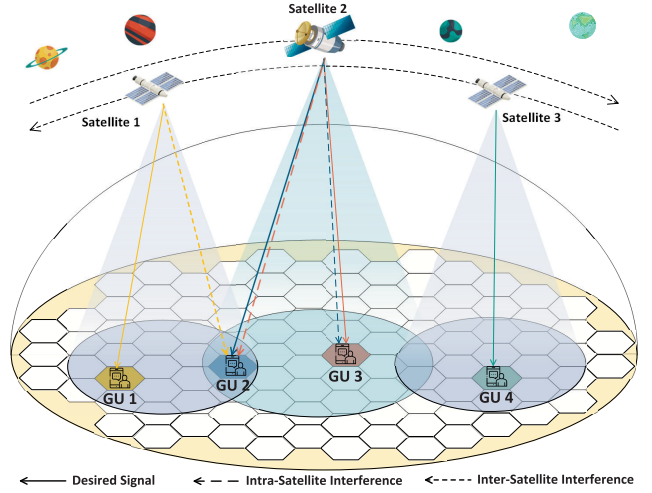


Fig. 1. Illustrative local-view schematic of the considered STIN hybrid beamforming scenario.

GWM-RA framework, including the graph-based world model architecture and latent-space policy optimization algorithm. Section IV analyzes the numerical results and compares the energy and sample efficiency of GWM-RA with the benchmarks. Finally, Section V concludes the paper.

II. SYSTEM MODEL AND PROBLEM FORMULATION

This section presents the mathematical model of the multi-satellite HBF system. We first define the network architecture and hybrid beamforming model. We then describe the channel, signal, and power-consumption models. Finally, we formulate the energy-efficiency maximization problem addressed by GWM-RA.

A. Network and Hybrid Beamforming Architecture

This work considers a downlink multibeam LEO satellite network. Fig. 1 illustrates a local view of the considered HBF scenario rather than the exact simulated topology. The numerical experiments use $S = 5$ cooperating satellites, as specified in Table II. The space segment comprises a constellation of S satellites, denoted by the set $\mathcal{S} = \{1, \dots, S\}$. These satellites provide service to K single-antenna ground users (GUs), collected in the set $\mathcal{K} = \{1, \dots, K\}$. Satellite trajectories and ephemerides are assumed to be known, so the visibility set $\mathcal{V}_k(t) \subseteq \mathcal{S}$ can be determined geometrically at each short slot t . Based on this visibility information and slowly varying traffic statistics, a higher-level scheduler determines the long-timescale scheduled-user set $\mathcal{K}_s[n] \subseteq \mathcal{K}$ and association matrix $\alpha[n] = [\alpha_{s,k}[n]]$, which remain fixed within interval n . Accordingly, this work focuses on short-slot digital precoding under fixed long-timescale topology management rather than joint real-time scheduling under arbitrary topology changes.

Each satellite is equipped with a uniform planar array (UPA) of $N_{\text{TX}} = N_x N_y$ antennas and N_{RF} radio frequency (RF) chains. A hybrid beamforming architecture is employed, which consists of an analog precoder, $\mathbf{F}_s^{\text{RF}}[n] \in \mathbb{C}^{N_{\text{TX}} \times N_{\text{RF}}}$, cascaded with a short-timescale digital baseband precoder, $\mathbf{F}_s^{\text{BB}}[t; n] \in \mathbb{C}^{N_{\text{RF}} \times |\mathcal{K}_s[n]|}$. The analog precoder is realized

TABLE II
SIMULATION PARAMETERS

Parameter	Symbol	Value
<i>Constellation and Scenario Parameters</i>		
Number of cooperating satellites	S	5
Satellite altitude	h	550 km
Orbital inclination		53°
Carrier frequency	f_c	20 GHz (Ka-band)
System bandwidth	B	100 MHz
User distribution		Uniform within coverage
<i>Channel and System Parameters</i>		
Noise power spectral density	N_0	-174 dBm/Hz
Rician K -factor	K	10 dB
Satellite antenna elements	N_{TX}	64 (8×8 UPA)
Satellite RF chains	N_{RF}	8
Number of phase shifters	N_{PS}	512 ($N_{\text{TX}} \times N_{\text{RF}}$)
<i>Power Model and Constraints</i>		
PA efficiency	η	0.5
Power per RF chain	P_0^{RF}	0.5 W
Power per phase shifter	P_0^{PS}	0.1 W
Static power consumption	P_c	10 W
Rate-dependent coefficient	c_{eff}	10 W/Gbps
Maximum satellite power	P_s^{max}	200 W
Minimum required rate	R^{min}	1 Mbps
<i>GWM-RA Algorithm Parameters</i>		
GNN layers		3
Optimizer		Adam
Learning rate		1×10^{-4}
Batch size		128
Replay buffer size		1×10^5

by a network of phase shifters, imposing a constant-modulus constraint on its elements, i.e., $[\mathbf{F}_s^{\text{RF}}]_{m,n} = 1/\sqrt{N_{\text{TX}}}$. The number of simultaneously served downlink streams per satellite is limited by the number of RF chains,

$$|\mathcal{K}_s[n]| \leq N_{\text{RF}}, \quad \forall s \in \mathcal{S}, \quad \forall n. \quad (1)$$

The system operates on two timescales. A long-timescale interval n of duration T_{long} contains short-timescale decision slots $t \in \mathcal{T}_n$, each of duration T_{short} . At the start of interval n , the analog precoders $\{\mathbf{F}_s^{\text{RF}}[n]\}$ and association decisions $\alpha[n]$ are fixed from slowly varying information, whereas the digital precoders $\{\mathbf{F}_s^{\text{BB}}[t;n]\}$ are updated at every short slot using instantaneous CSI. The interval index $[n]$ is omitted later when the long-timescale context is clear.

The analog stage is configured using a codebook-based procedure that satisfies the constant-modulus hardware constraint, as described in Section III-A.

B. Channel Model

This subsection describes the downlink channel from satellite $s \in \mathcal{S}$ to user $k \in \mathcal{K}$. Transmission is organized in short-timescale slots indexed by t . The instantaneous channel is constant within each slot of duration T_{short} and varies independently across slots, while the large-scale components evolve over the long-timescale interval n of duration T_{long} due to satellite mobility and tropospheric propagation.

The attenuation is modeled as the sum of free-space path loss (FSPL) and an elevation-dependent atmospheric component accounting for gaseous absorption, rain attenuation, and

cloud losses in accordance with ITU-R recommendations [29]. For a slant range $d_{s,k}$, elevation $\theta_{s,k}$, carrier frequency f , and speed of light c , the total path loss is

$$L_{s,k} = 20 \log_{10} \left(\frac{4\pi f d_{s,k}}{c} \right) + L_{\text{atm}}(\theta_{s,k}, f), \quad (2)$$

where $L_{\text{atm}}(\theta_{s,k}, f)$ denotes the atmospheric attenuation component at elevation angle $\theta_{s,k}$ and carrier frequency f . After converting to linear scale and incorporating antenna gains, the large-scale power gain is

$$\beta_{s,k} = G_s^{\text{TX}} G_k^{\text{RX}} \cdot 10^{-L_{s,k}/10}, \quad (3)$$

where G_s^{TX} and G_k^{RX} denote the transmit antenna gain of satellite s and the receive antenna gain of user k , respectively.

The small-scale fading characteristics of the LEO satellite channel are captured by a slot-wise Rician block-fading model. The normalized small-scale channel vector in short slot t is written as

$$\tilde{\mathbf{h}}_{s,k}[t] = \sqrt{\frac{K_{s,k}}{K_{s,k} + 1}} \mathbf{a}(\theta_{s,k}, \phi_{s,k}) e^{j\varphi_{s,k}[t]} + \sqrt{\frac{1}{K_{s,k} + 1}} \tilde{\mathbf{g}}_{s,k}[t], \quad (4)$$

where $K_{s,k}$ is the Rician factor, $\varphi_{s,k}[t] \sim \text{Unif}[0, 2\pi)$ is the slot-wise LoS phase, and $\tilde{\mathbf{g}}_{s,k}[t] \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_{N_{\text{TX}}})$ represents the NLoS scattering component. The random variables $\varphi_{s,k}[t]$ and $\tilde{\mathbf{g}}_{s,k}[t]$ are independently drawn across short-timescale slots under the adopted i.i.d. block-fading abstraction. For a UPA with element spacings d_x and d_y , the steering vector is given by

$$\mathbf{a}(\theta, \phi) = \mathbf{a}_x(\theta, \phi) \otimes \mathbf{a}_y(\theta, \phi), \quad (5)$$

where θ and ϕ are the elevation and azimuth angles, respectively. The components of the steering vectors $\mathbf{a}_x \in \mathbb{C}^{N_x}$ and $\mathbf{a}_y \in \mathbb{C}^{N_y}$ are

$$[\mathbf{a}_x(\theta, \phi)]_{n_x} = e^{j\pi(n_x-1)\frac{d_x}{\lambda} \sin \phi \cos \theta},$$

$$[\mathbf{a}_y(\theta, \phi)]_{n_y} = e^{j\pi(n_y-1)\frac{d_y}{\lambda} \cos \phi},$$

with λ being the carrier wavelength. Combining the large-scale gain and small-scale fading gives the channel vector from satellite s to user k as

$$\mathbf{h}_{s,k}[t] = \sqrt{\beta_{s,k}} \tilde{\mathbf{h}}_{s,k}[t]. \quad (6)$$

The small-scale fading across different short-timescale slots, as well as the interference-link channels, is modeled as i.i.d. under a block-fading abstraction. This supports the horizon-one episodic RL formulation adopted later. The assumption is appropriate when each short-timescale decision slot can be treated as one coherence block and successive training samples are taken from independently regenerated channel realizations. In scenarios with strong temporal correlation, sequence-based extensions would be required.

C. Signal Model and SINR

For a given satellite s , let the data symbols intended for its scheduled users $\mathcal{K}_s$ be collected in a vector $\mathbf{d}_s = [d_k]_{k \in \mathcal{K}_s}$, where each symbol d_k is drawn from a complex Gaussian distribution with $\mathbb{E}[|d_k|^2] = 1$. The digital baseband precoder is a matrix $\mathbf{F}_s^{\text{BB}} = [\mathbf{f}_k^{\text{BB}}]_{k \in \mathcal{K}_s} \in \mathbb{C}^{N_{\text{RF}} \times |\mathcal{K}_s|}$, where the vector $\mathbf{f}_k^{\text{BB}}$ is the digital precoder for user k and its norm determines the allocated power, i.e., $\|\mathbf{f}_k^{\text{BB}}\|_2^2 = p_k$. Consequently, the effective hybrid precoding vector for user k is given by

$$\mathbf{w}_k = \mathbf{F}_s^{\text{RF}} \mathbf{f}_k^{\text{BB}}. \quad (7)$$

The aggregate signal transmitted from satellite s is the superposition of all user streams, given by $\mathbf{x}_s = \sum_{k \in \mathcal{K}_s} \mathbf{w}_k d_k$. Let s_k^* denote the satellite serving user k . The received signal at user k is then expressed as

$$\begin{aligned} y_k = & \underbrace{\mathbf{h}_{s_k^*,k}^H \mathbf{w}_k d_k}_{\text{desired signal}} + \underbrace{\sum_{\substack{j \in \mathcal{K}_{s_k^*} \\ j \neq k}} \mathbf{h}_{s_k^*,k}^H \mathbf{w}_j d_j}_{\text{intra-satellite interference}} \\ & + \underbrace{\sum_{\substack{s \in \mathcal{S} \\ s \neq s_k^*}} \sum_{j \in \mathcal{K}_s} \mathbf{h}_{s,k}^H \mathbf{w}_j d_j}_{\text{inter-satellite interference}} + n_k, \end{aligned} \quad (8)$$

where the second term on the right-hand side represents the intra-satellite interference, the third term is the inter-satellite interference, and $n_k \sim \mathcal{CN}(0, \sigma^2)$ is the additive white Gaussian noise (AWGN) with variance σ^2 .

The aggregate interference power experienced by user k is defined as

$$I_k = \sum_{\substack{j \in \mathcal{K}_{s_k^*} \\ j \neq k}} |\mathbf{h}_{s_k^*,k}^H \mathbf{w}_j|^2 + \sum_{\substack{s \in \mathcal{S} \\ s \neq s_k^*}} \sum_{j \in \mathcal{K}_s} |\mathbf{h}_{s,k}^H \mathbf{w}_j|^2. \quad (9)$$

The resulting signal-to-interference-plus-noise ratio (SINR) for user k is

$$\Gamma_k = \frac{|\mathbf{h}_{s_k^*,k}^H \mathbf{w}_k|^2}{I_k + \sigma^2}. \quad (10)$$

Assuming Gaussian signaling, the achievable data rate for user k (in bits/s) is given by the Shannon capacity formula:

$$R_k = B \log_2(1 + \Gamma_k), \quad (11)$$

where B denotes the system bandwidth.

Remarks: (i) If orthogonal user scheduling is adopted on a satellite (e.g., one stream per slot), the intra-satellite interference term vanishes; the present model allows more general non-orthogonal multiplexing. (ii) The total transmit power of satellite s in a slot is $\sum_{k \in \mathcal{K}_s} \|\mathbf{w}_k\|_2^2$, which will be used in the power-consumption model and the ensuing optimization.

D. Power Consumption Model

The total power consumption of each satellite $s \in \mathcal{S}$ includes both transmission-dependent and hardware-related components, as detailed below.

1) *Transmit Power and Amplifier Efficiency:* For satellite $s \in \mathcal{S}$, the total radiated power is determined by the effective precoding vectors of its scheduled users. Specifically,

$$P_s^{\text{TX}} = \sum_{k \in \mathcal{K}_s} \|\mathbf{w}_k\|_2^2, \quad (12)$$

where $\mathbf{w}_k$ is the effective hybrid precoder of user k . Due to the limited efficiency $\eta \in (0, 1]$ of power amplifiers (PAs), the actual PA power consumption exceeds the radiated power, and is modeled as

$$P_s^{\text{PA}} = \frac{1}{\eta} P_s^{\text{TX}}. \quad (13)$$

2) *RF Chains and Phase Shifters:* The hybrid beamforming architecture employs multiple RF chains and phase shifters. If each satellite is equipped with N_{RF} RF chains and N_{PS} phase shifters, their static power consumptions are given by

$$P_s^{\text{RF}} = N_{\text{RF}} P_0^{\text{RF}}, \quad P_s^{\text{PS}} = N_{\text{PS}} P_0^{\text{PS}}, \quad (14)$$

where P_0^{RF} and P_0^{PS} denote the power per RF chain and per phase shifter, respectively. These terms represent constant hardware overhead independent of instantaneous scheduling.

3) *Baseband Processing and Rate-Dependent Power:* In addition to the RF hardware, satellites consume power for baseband operations, channel coding/decoding, and backhaul forwarding. This consumption is modeled as a constant circuit term plus a rate-dependent component:

$$P_s^{\text{BB}} = P_c + c_{\text{eff}} \sum_{k \in \mathcal{K}_s} R_k, \quad (15)$$

where P_c captures constant baseband and control-plane power, and c_{eff} is a proportional coefficient (converted to W/bps from the nominal W/Gbps specification) accounting for rate-dependent computation and backhaul load.

By summing the above contributions, the total power consumption of satellite s is

$$P_s^{\text{tot}} = P_s^{\text{PA}} + P_s^{\text{RF}} + P_s^{\text{PS}} + P_s^{\text{BB}}. \quad (16)$$

Aggregating over the entire constellation, system-level energy efficiency (EE) is defined as

$$\text{EE} = \frac{\sum_{k \in \mathcal{K}} R_k}{\sum_{s \in \mathcal{S}} P_s^{\text{tot}}} \quad [\text{bits/Joule}]. \quad (17)$$

For clarity in numerical evaluations, this metric is reported in Mbps/W, which is dimensionally equivalent to Megabits per Joule (Mbits/J).

This metric captures the tradeoff between achievable throughput and power expenditure and will serve as the optimization objective in the following formulation. Note that P_s^{tot} depends implicitly on both analog and digital precoders through the transmit power term.

E. Problem Formulation

Let $\mathcal{T}_n$ denote the set of short-timescale slots contained in long-timescale interval n . The overall mixed-timescale design objective is to maximize the average system-wide energy efficiency within each interval by jointly optimizing the long-timescale analog/scheduling variables and the short-timescale digital precoders. For convenience, define the scheduled-user

set in long-timescale interval n as $\mathcal{K}^{\text{sch}}[n] = \bigcup_{s \in \mathcal{S}} \mathcal{K}_s[n]$, and collect all joint design variables in $\mathcal{X}[n]$. The resulting problem is formulated as

$$\begin{aligned}
 \text{(P0)} \quad & \underset{\mathcal{X}[n]}{\text{maximize}} \quad \frac{1}{|\mathcal{T}_n|} \sum_{t \in \mathcal{T}_n} \text{EE}[t; n] \\
 \text{s.t.} \quad & \text{C1: } R_k[t; n] \geq R^{\min}, \\
 & \quad \forall (k, t) \in \mathcal{K}^{\text{sch}}[n] \times \mathcal{T}_n, \\
 & \text{C2: } P_s^{\text{tot}}[t; n] \leq P_s^{\max}, \\
 & \quad \forall (s, t) \in \mathcal{S} \times \mathcal{T}_n, \\
 & \text{C3: } \|\mathbf{f}_k^{\text{BB}}[t; n]\|_2^2 \leq p_k^{\max}, \\
 & \quad \forall (k, t) \in \mathcal{K}^{\text{sch}}[n] \times \mathcal{T}_n, \\
 & \text{C4: } \mathbf{F}_s^{\text{RF}}[n] \in \mathcal{F}_s, \quad \forall s \in \mathcal{S}, \\
 & \text{C5: } \alpha_{s,k}[n] \in \{0, 1\}, \\
 & \quad \forall (s, k) \in \mathcal{S} \times \mathcal{K}, \\
 & \text{C6: } \sum_{k \in \mathcal{K}} \alpha_{s,k}[n] \leq N_{\text{RF}}, \\
 & \quad \forall s \in \mathcal{S}, \\
 & \text{C7: } \sum_{s \in \mathcal{S}} \alpha_{s,k}[n] \leq 1, \forall k \in \mathcal{K}. \quad (18)
 \end{aligned}$$

Here, $\mathcal{F}_s$ denotes the feasible set of analog precoders for satellite s , i.e.,

$$\mathcal{F}_s = \left\{ \mathbf{F} \in \mathbb{C}^{N_{\text{TX}} \times N_{\text{RF}}} : \|\mathbf{F}\|_{m,n} = N_{\text{TX}}^{-1/2} \right\}. \quad (19)$$

Problem (P0) describes the overall mixed-timescale design objective. It is a large-scale non-convex mixed-integer fractional program whose long-timescale variables $\{\mathbf{F}_s^{\text{RF}}[n]\}$ and $\alpha[n]$ couple with the short-timescale variables $\{\mathbf{f}_k^{\text{BB}}[t; n]\}_{t \in \mathcal{T}_n}$. Obtaining a joint global solution to (P0) is computationally intractable.

In this paper, the long-timescale variables are assumed to be configured by a higher-level scheduler from slowly varying network information. Conditioned on the fixed long-timescale decisions $\{\mathbf{F}_s^{\text{RF}}[n]\}$ and $\alpha[n]$, let $\mathbf{F}^{\text{BB}}[t; n]$ collect the digital precoders of all scheduled users. The short-slot digital precoding problem solved by the proposed GWM-RA at each $t \in \mathcal{T}_n$ is

$$\begin{aligned}
 \text{(P1)} \quad & \underset{\mathbf{F}^{\text{BB}}[t; n]}{\text{maximize}} \quad \text{EE}[t; n] \\
 \text{s.t.} \quad & \text{C1: } R_k[t; n] \geq R^{\min}, \quad \forall k \in \mathcal{K}^{\text{sch}}[n], \\
 & \text{C2: } P_s^{\text{tot}}[t; n] \leq P_s^{\max}, \quad \forall s \in \mathcal{S}, \\
 & \text{C3: } \|\mathbf{f}_k^{\text{BB}}[t; n]\|_2^2 \leq p_k^{\max}, \\
 & \quad \forall k \in \mathcal{K}^{\text{sch}}[n]. \quad (20)
 \end{aligned}$$

Although (P1) is simpler than (P0), it remains non-convex because of coupled interference terms. We therefore learn a non-iterative policy that maps each short-slot channel realization directly to a digital precoding action. Under the i.i.d. block-fading assumption, this is naturally posed as a horizon-one episodic RL problem, equivalently a contextual decision problem, with tuple $(\mathcal{S}, \mathcal{A}, R)$ defined below:

- State Space $\mathcal{S}$: $s_t = \{\mathbf{h}_{s,k}[t; n] \mid s \in \mathcal{S}, k \in \mathcal{K}\}$.
- Action Space $\mathcal{A}$: $a_t = \{\mathbf{f}_k^{\text{BB}}[t; n]\}_{k \in \bigcup_s \mathcal{K}_s[n]}$, since association is fixed by the higher-level scheduler.

- Reward Function R : $r_t = R(s_t, a_t) = \text{EE}[t; n]$.

For brevity, the interval index $[n]$ is omitted in Section III when the long-timescale context is clear.

III. PROPOSED METHOD

This section presents the proposed GWM-RA for the short-slot digital precoding problem (P1) under fixed long-timescale analog precoders and scheduler-provided association decisions. We first describe the long-timescale analog beam design procedure. We then present the graph-conditioned world model and the actor-critic policy-learning procedure for short-slot digital precoding. Finally, we discuss the computational characteristics of the resulting framework.

A. Two-Stage Hybrid Beamforming

The proposed method follows the two-timescale operation defined in Section II-A. This subsection describes the long-timescale configuration of the analog precoder $\mathbf{F}_s^{\text{RF}}$ based on statistical channel properties derived from the geographical user distribution. The candidate beams are drawn from a standard two-dimensional discrete Fourier transform (DFT) codebook $\mathbf{D} \in \mathbb{C}^{N_{\text{TX}} \times N_{\text{TX}}}$ [30]. The codebook is written as $\mathbf{D} = [\mathbf{d}_1, \dots, \mathbf{d}_{N_{\text{TX}}}]$, where each column is a steering vector constructed as the Kronecker product $\mathbf{d} = \mathbf{d}_y \otimes \mathbf{d}_x$, with elements

$$\begin{aligned}
 [\mathbf{d}_x]_{n_x} &= \frac{1}{\sqrt{N_x}} e^{j\pi \frac{d_x}{\lambda} (n_x - 1) \sin(\theta_l^*)}, \quad n_x = 1, \dots, N_x, \\
 [\mathbf{d}_y]_{n_y} &= \frac{1}{\sqrt{N_y}} e^{j\pi \frac{d_y}{\lambda} (n_y - 1) \sin(\phi_m^*)}, \quad n_y = 1, \dots, N_y.
 \end{aligned} \quad (21)$$

Here, $\sin(\theta_l^*) = 1 - \frac{2}{N_x} l$, $l = 0, 1, \dots, N_x - 1$, and $\sin(\phi_m^*) = 1 - \frac{2}{N_y} m$, $m = 0, 1, \dots, N_y - 1$.

Let $\mathcal{D} = \{\mathbf{d}_m\}_{m=1}^{N_{\text{TX}}}$ denote the set of all candidate codeword vectors contained in $\mathbf{D}$. Given the codebook set $\mathcal{D}$, the analog precoder $\mathbf{F}_s^{\text{RF}}$ is constructed by selecting an N_{RF} -sized subset of codewords that provides effective coverage for the scheduled users $\mathcal{K}_s$. The analog precoder is constructed using a two-step greedy subset-selection procedure. First, the spatial signature of each scheduled user $k \in \mathcal{K}_s$ is mapped to a codeword in $\mathcal{D}$. The primary candidate $\mathbf{d}_k^*$ maximizes the array gain for the line-of-sight (LoS) angles $(\theta_{s_k^*,k}, \phi_{s_k^*,k})$:

$$\mathbf{d}_k^* = \arg \max_{\mathbf{d} \in \mathcal{D}} |\mathbf{a}(\theta_{s_k^*,k}, \phi_{s_k^*,k})^H \mathbf{d}|^2. \quad (22)$$

The resulting candidate set is $\mathcal{D}_s^* = \{\mathbf{d}_k^*\}_{k \in \mathcal{K}_s}$. A greedy algorithm then selects codewords to satisfy the RF-chain dimensionality constraint while providing broad spatial coverage. Let $\mathcal{S}^{(i)}$ denote the ordered set of selected codewords at iteration i , and let $\mathcal{U}^{(i)}$ denote the corresponding set of covered users, evaluated using the array-gain threshold γ_{th} . At each iteration, the algorithm selects the candidate in $\mathcal{D}_s^*$ that covers the largest number of additional users while satisfying the minimum angular-separation threshold $\theta_{\min}$ relative to the codewords in $\mathcal{S}^{(i-1)}$. The procedure is summarized in Algorithm 1.

Algorithm 1 Topology-Aware Greedy Analog Precoder Configuration

Input: $\mathcal{K}_s, \mathcal{D}, \{\mathbf{a}_k\}, N_{\text{RF}}, \theta_{\min}$, and γ_{th} .
Output: Analog precoding matrix $\mathbf{F}_s^{\text{RF}}$.

```

1 Define  $\delta(\mathbf{d}, \mathcal{S}) = \min_{\mathbf{v} \in \mathcal{S}} \arccos(|\mathbf{d}^H \mathbf{v}| / (\|\mathbf{d}\| \|\mathbf{v}\|))$ ;
2 Define  $\mathcal{C}(\mathbf{d}, \mathcal{U}) = \{k \in \mathcal{K}_s \setminus \mathcal{U} : |\mathbf{a}_k^H \mathbf{d}|^2 \geq \gamma_{\text{th}}\}$ ;
3  $\mathcal{D}_s^* \leftarrow \emptyset, \mathcal{S}^{(0)} \leftarrow \emptyset$ , and  $\mathcal{U}^{(0)} \leftarrow \emptyset$ ;
4 for each user  $k \in \mathcal{K}_s$  do
5    $\mathbf{d}_k^* \leftarrow \arg \max_{\mathbf{d} \in \mathcal{D}} |\mathbf{a}_k^H \mathbf{d}|^2$ ;
6    $\mathcal{D}_s^* \leftarrow \mathcal{D}_s^* \cup \{\mathbf{d}_k^*\}$ ;
7 end
8 for iteration  $i = 1, \dots, N_{\text{RF}}$  do
9   if  $i = 1$  then
10     $\mathcal{F}^{(i)} \leftarrow \mathcal{D}_s^*$ ;
11   else
12     $\mathcal{F}^{(i)} \leftarrow \{\mathbf{d} \in \mathcal{D}_s^* \setminus \mathcal{S}^{(i-1)} : \delta(\mathbf{d}, \mathcal{S}^{(i-1)}) \geq \theta_{\min}\}$ ;
13   end
14   if  $\mathcal{F}^{(i)} \neq \emptyset$  then
15     $\mathbf{d}^{(i)} \leftarrow \arg \max_{\mathbf{d} \in \mathcal{F}^{(i)}} |\mathcal{C}(\mathbf{d}, \mathcal{U}^{(i-1)})|$ ;
16   else
17    Randomly select  $\mathbf{d}^{(i)} \in \mathcal{D} \setminus \mathcal{S}^{(i-1)}$ ;
18   end
19    $\mathcal{S}^{(i)} \leftarrow \mathcal{S}^{(i-1)} \cup \{\mathbf{d}^{(i)}\}$ ;
20    $\mathcal{U}^{(i)} \leftarrow \mathcal{U}^{(i-1)} \cup \mathcal{C}(\mathbf{d}^{(i)}, \emptyset)$ ;
21 end
22  $\mathbf{F}_s^{\text{RF}} \leftarrow [\mathbf{d}^{(1)}, \dots, \mathbf{d}^{(N_{\text{RF}})}]$ ;
23 return  $\mathbf{F}_s^{\text{RF}}$ 

```

This analog precoder is updated on the long timescale T_{long} with complexity dominated by user-to-codeword matching, i.e., $\mathcal{O}(|\mathcal{K}_s|N_{\text{TX}}^2)$. The resulting $\mathbf{F}_s^{\text{RF}}$ satisfies the constant-modulus constraint by construction and remains fixed during short-timescale operation.

B. GWM-RA for Digital Precoder Optimization

This subsection presents GWM-RA for the short-slot digital precoding problem (P1) under fixed long-timescale analog precoders and scheduler-provided association decisions. The action space is therefore aligned with the digital-only scope of (P1). However, the learned surrogate used for policy optimization is not fully equivalent to (P1): while user association is fixed and the digital precoder is optimized at each short slot, the per-satellite power constraint C2 is handled through an aggregate relaxation in the imagined reward, whereas exact per-satellite feasibility is evaluated only in the environment. For analytical tractability and explicit action-level feasibility, the actor outputs unconstrained real variables $\tilde{p}_k \in \mathbb{R}$ and $\tilde{\mathbf{v}}_k^{(R)}, \tilde{\mathbf{v}}_k^{(I)} \in \mathbb{R}^{N_{\text{RF}}}$ for each scheduled user $k \in \mathcal{U}_s \mathcal{K}_s[n]$. Define the complex direction prototype $\tilde{\mathbf{v}}_k = \tilde{\mathbf{v}}_k^{(R)} + j\tilde{\mathbf{v}}_k^{(I)}$, and map the raw actor outputs to the deployed digital precoder via

$$\tilde{\mathbf{v}}_k = \tilde{\mathbf{v}}_k^{(R)} + j\tilde{\mathbf{v}}_k^{(I)}, \quad p_k = p_k^{\max} \sigma(\tilde{p}_k),$$

$$\mathbf{u}_k = \frac{\tilde{\mathbf{v}}_k}{\max(\|\tilde{\mathbf{v}}_k\|_2, \epsilon)}, \quad \mathbf{f}_k^{\text{BB}} = \sqrt{p_k} \mathbf{u}_k, \quad (23)$$

where $\sigma(\cdot)$ denotes the sigmoid function and $\epsilon > 0$ is a small constant. Hence $\|\mathbf{u}_k\|_2 \leq 1$ and $\|\mathbf{f}_k^{\text{BB}}\|_2^2 \leq p_k \leq p_k^{\max}$, so constraint C3 is satisfied by construction. Accordingly, the proposed policy should be interpreted as digital-only at the decision-variable level, while constraint C2 is still handled approximately in the surrogate model through the

aggregate power penalty and is checked exactly only in the true environment.

The correspondence between GWM-RA and the horizon-one RL formulation of (P1) is summarized as follows:

- **State** s_t : a heterogeneous graph $\mathcal{G}_t$, encoded by the GNN into $\mathbf{z}_t$.
- **Action** a_t : the scheduled users' digital-precoding vectors $\{\mathbf{f}_k^{\text{BB}}\}_{k \in \mathcal{U}_s \mathcal{K}_s[n]}$. For parameterization, each digital precoder is written as

$$\mathbf{f}_k^{\text{BB}} = \sqrt{p_k} \mathbf{u}_k, \quad 0 \leq p_k \leq p_k^{\max}, \quad \|\mathbf{u}_k\|_2 \leq 1. \quad (24)$$

- **Reward** r_t : the true short-slot system EE, i.e., $r_t = \text{EE}[t; n]$.

In GWM-RA, the world model represents the task-relevant performance of a digital-precoding action under the current graph state. Under the adopted i.i.d. block-fading model, the next short-slot channel realization is independent of the current action. Thus, learning $p(s_{t+1}|s_t, a_t)$ or recurrent latent dynamics would characterize action-independent channel variation rather than the action-dependent signal needed to optimize the current-slot precoder. We therefore model the immediate graph-conditioned action–performance mapping

$$f_{\text{WM}} : (\mathcal{G}_t, a_t) \mapsto (\hat{\mathbf{R}}_t, \hat{P}_t), \quad (25)$$

where $\mathcal{G}_t$ represents the current graph-structured CSI and interference state, a_t denotes a candidate digital-precoding action, $\hat{\mathbf{R}}_t$ is the predicted user-wise rate vector, and $\hat{P}_t$ is the predicted total power consumption. The model is trained in a supervised manner using environment-generated labels $(\mathbf{R}_t, P_t)$ obtained from true rate and power evaluations. The latent vector $\mathbf{z}_t$ produced by the GNN encoder serves as an intermediate graph representation for the prediction heads and is not the supervised prediction target of the world model. After training, f_{WM} provides differentiable one-step reward and constraint estimates for actor/critic updates, reducing repeated true-environment reward queries during policy improvement. The present implementation uses this one-step surrogate evaluation and does not include recurrent latent dynamics or multi-step rollouts.

1) *Graph-Based World Model*: A GNN is adopted because the satellite–user links and their interference couplings form a dynamic heterogeneous graph. Unlike MLPs, which are sensitive to varying node counts and do not explicitly represent structural dependencies, GNN message passing provides an inductive bias for modeling non-local interference within the considered topology family.

The world model learns a differentiable mapping from the graph-structured state s_t and action a_t to the user-wise rate vector $\hat{\mathbf{R}}_t = [\hat{R}_k]_{k \in \mathcal{U}_s \mathcal{K}_s[n]}$ and total power $\hat{P}_t$, from which $\hat{R}_t = \sum_{k \in \mathcal{U}_s \mathcal{K}_s[n]} \hat{R}_k$ is recovered. It comprises a GNN-based state encoder and two MLP prediction heads.

State Encoder: In each time slot t , the physical state $s_t = \{\mathbf{h}_{s,k}[t]\}$ is represented as a heterogeneous graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E})$. The vertex set $\mathcal{V}$ contains satellite and user nodes, with node features representing their properties. The edge set $\mathcal{E}$ represents potential communication and interference

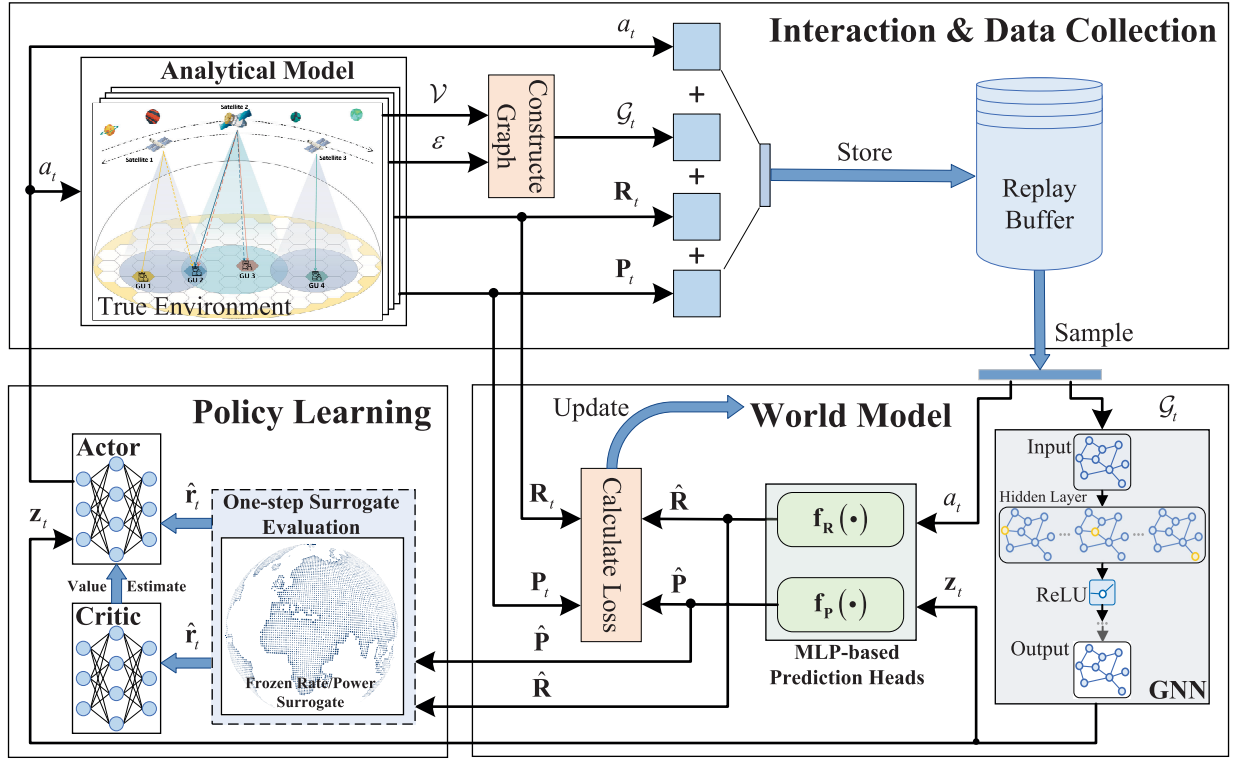


Fig. 2. Implemented training pipeline of GWM-RA under i.i.d. block fading. The world model is a graph-conditioned, action-conditioned performance surrogate for user-wise rates and total power, $f_{WM} : (\mathcal{G}_t, a_t) \mapsto (\hat{\mathbf{R}}_t, \hat{\mathbf{P}}_t)$, and its one-step outputs are used for actor-critic updates. The implementation does not include a recurrent transition module or multi-step rollout.

links, with edge features defined by the corresponding channel vectors.

A GNN with L message-passing layers encodes $\mathcal{G}_t$ into a low-dimensional latent state vector $\mathbf{z}_t$ that represents the network interference topology. At each layer $l \in \{1, \dots, L\}$, message passing is performed using a heterogeneous attention mechanism. For node $u \in \mathcal{V}$, the unnormalized attention score e_{vu} from neighboring node $v \in \mathcal{N}_u$ is

$$e_{vu} = \text{LeakyReLU} \left(\mathbf{a}_{\text{rel}}^T [\mathbf{W}_u \mathbf{x}_u^{(l)} \parallel \mathbf{W}_v \mathbf{x}_v^{(l)} \parallel \mathbf{W}_e \mathbf{x}_{vu}] \right). \quad (26)$$

Here, $\mathbf{x}_u^{(l)}$ is the feature vector of node u at layer l , $\mathbf{x}_{vu}$ is the feature of the edge from v to u , $[\cdot \parallel \cdot]$ denotes concatenation, and $\{\mathbf{W}_u, \mathbf{W}_v, \mathbf{W}_e, \mathbf{a}_{\text{rel}}\}$ are relation-specific learnable weight matrices and an attention vector.

The attention scores are then normalized using the softmax function to obtain the final coefficients:

$$\alpha_{vu} = \frac{\exp(e_{vu})}{\sum_{j \in \mathcal{N}_u} \exp(e_{ju})}. \quad (27)$$

The feature vector of node u for the next layer, $\mathbf{x}_u^{(l+1)}$, is updated by aggregating information from its neighbors, weighted by the attention coefficients:

$$\mathbf{x}_u^{(l+1)} = \sigma \left(\sum_{v \in \mathcal{N}_u} \alpha_{vu} \mathbf{W}_v \mathbf{x}_v^{(l)} \right), \quad (28)$$

where $\sigma(\cdot)$ is a nonlinear activation function. After L layers, a readout function aggregates the final node features $\{\mathbf{x}_u^{(L)}\}$ to produce the graph-level latent state vector $\mathbf{z}_t$.

Fig. 2 illustrates the GWM-RA training pipeline implemented in this paper. While standard world models often include a recurrent component to predict temporal state transitions in sequential tasks, the physical channel states in our LEO network scenario are modeled as independent and identically distributed (i.i.d.) across short-timescale slots under the block fading assumption. Unlike physical control tasks, the beamforming action a_t taken in slot t does not alter the subsequent physical channel realization s_{t+1} .

Accordingly, under the considered horizon-one i.i.d. block-fading setting, the world-model component is a graph-conditioned surrogate of the task-relevant performance quantities, namely the user-wise rate vector and total power consumption, from which the imagined reward is constructed.

Performance Prediction Heads: The latent state $\mathbf{z}_t$ is concatenated with the parameterized action a_t and passed to two separate multilayer-perceptron (MLP) prediction heads:

- **Rate Predictor (f_R):** This head outputs the predicted user-wise rate vector, $\hat{\mathbf{R}}_t = f_R(\mathbf{z}_t, a_t)$, whose k -th component is denoted by $\hat{R}_k$.
- **Power Predictor (f_P):** This head outputs the predicted total power consumption, $\hat{P}_t = f_P(\mathbf{z}_t, a_t)$.

World Model Training: The world model is trained by minimizing a weighted mean-square error (MSE) between its predictions and the corresponding quantities obtained from environment interactions. Let $\Theta_{WM} = \{\Theta_{\text{enc}}, \Theta_R, \Theta_P\}$ denote the trainable parameters of the GNN encoder, rate predictor, and power predictor, respectively. Using experience

tuples $(s_t, a_t, \mathbf{R}_t, P_t)$ sampled from the replay buffer $\mathcal{B}$, where $\mathbf{R}_t = [R_k]_{k \in \mathcal{K}^{\text{sch}}[n]}$ is the true user-wise rate vector, the loss is

$$\min_{\Theta_{WM}} \mathcal{L}_{WM}(\Theta_{WM}) = \mathbb{E}_{(s_t, a_t, \mathbf{R}_t, P_t) \sim \mathcal{B}} \left[\omega_R \|\hat{\mathbf{R}}_t - \mathbf{R}_t\|_2^2 + \omega_P (\hat{P}_t - P_t)^2 \right], \quad (29)$$

where ω_R and ω_P are weighting coefficients for the user-wise rate prediction and the power prediction, respectively.

2) *Policy Learning*: Once the world model is trained, the resource-allocation policy is learned using an actor-critic method. Policy learning operates in the latent space and uses the world model as a differentiable surrogate of the task-relevant system performance. The actor and critic are trained from an imagined reward $\hat{r}_t$ constructed from the world-model predictions. Because the current power predictor outputs only the aggregate power $\hat{P}_t$ rather than a per-satellite power vector, the power-constraint term in the imagined reward is implemented as an aggregate relaxation of C2. Specifically, the penalty is imposed relative to the sum power budget $\sum_{s \in \mathcal{S}} P_s^{\text{max}}$, while exact per-satellite feasibility is checked only in the true environment. Accordingly, the surrogate objective used for policy learning should be interpreted as a relaxed one-step proxy of (P1), not as an exact hard-constrained solver of (P1).

Unless otherwise stated, all performance results reported in Section IV are computed in the true environment from the actions actually executed by each method, without post-hoc projection, repair, or infeasibility filtering. The empirical QoS and power violation statistics of GWM-RA are reported in Section IV-D for the ablation operating point to quantify the practical effect of the penalty-augmented reward.

The imagined reward is defined as

$$\hat{r}_t = \frac{\sum_{k \in \mathcal{K}^{\text{sch}}[n]} \hat{R}_k}{\hat{P}_t} - \lambda_R^{\text{pen}} \sum_{k \in \mathcal{K}^{\text{sch}}[n]} \max(0, R^{\text{min}} - \hat{R}_k) - \lambda_P^{\text{pen}} \max\left(0, \hat{P}_t - \sum_{s \in \mathcal{S}} P_s^{\text{max}}\right). \quad (30)$$

Here, $\hat{\mathbf{R}}_t = f_R(\mathbf{z}_t, a_t)$ and $\hat{P}_t = f_P(\mathbf{z}_t, a_t)$ denote the predicted user-wise rate vector and the predicted aggregate power, respectively, while $\lambda_R^{\text{pen}}, \lambda_P^{\text{pen}} > 0$ are penalty coefficients.

The critic is a value network $V_\phi(\mathbf{z}_t)$, parameterized by ϕ , that estimates the expected one-step imagined reward for latent state $\mathbf{z}_t$ under the horizon-one formulation. It is trained by minimizing the mean squared error between its prediction and the imagined reward using data sampled from the replay buffer $\mathcal{B}$:

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{z}_t \sim \mathcal{B}} \left[(V_\phi(\mathbf{z}_t) - \hat{r}_t)^2 \right]. \quad (31)$$

The actor is a policy network $\pi_\theta(a_t|\mathbf{z}_t)$, parameterized by θ , that maps latent state $\mathbf{z}_t$ to a distribution over the action space. For the continuous digital-precoding action, this distribution is modeled as a multivariate Gaussian. The actor is updated using the policy-gradient theorem to maximize the expected imagined reward. The advantage function A_t is used to reduce variance:

$$A_t = \hat{r}_t - V_\phi(\mathbf{z}_t). \quad (32)$$

The policy objective is $J(\theta) = \mathbb{E}_{\mathbf{z}_t \sim \mathcal{B}, a_t \sim \pi_\theta} [\hat{r}_t]$. Its gradient is estimated using the advantage baseline as

$$\nabla_\theta J(\theta) = \mathbb{E}_{\substack{\mathbf{z}_t \sim \mathcal{B}, \\ a_t \sim \pi_\theta}} [\nabla_\theta \log \pi_\theta(a_t|\mathbf{z}_t) A_t]. \quad (33)$$

This surrogate-assisted procedure uses one-step reward estimates for actor-critic updates, thereby reducing repeated true-environment reward queries under the adopted training protocol.

3) *Training Protocol and Inference*: The GWM-RA procedure comprises offline training and online inference stages.

a) *Offline training protocol*: Training is off-policy. A replay buffer $\mathcal{B}$ is first populated with tuples $(s_t, a_t, \mathbf{R}_t, P_t)$ from random exploration in the true environment, where $\mathbf{R}_t = [R_k]_{k \in \mathcal{K}^{\text{sch}}[n]}$. Each training iteration then alternates between (i) updating Θ_{WM} by minimizing $\mathcal{L}_{WM}(\Theta_{WM})$ on mini-batches from $\mathcal{B}$ and (ii) freezing the world model and updating the actor and critic in latent space using encoded states $\mathbf{z}_t$ and imagined rewards $\hat{r}_t$. New experiences collected by the updated policy are appended to $\mathcal{B}$. This training is conditioned on a given long-timescale analog precoder configuration. Accordingly, the learned policy and world model do not claim zero-shot generalization to entirely new analog subspaces; substantial long-timescale reconfiguration may induce distribution shift and require lightweight fine-tuning or partial retraining.

b) *Online inference*: During deployment, a channel realization s_t is encoded into $\mathbf{z}_t$, and the actor outputs the digital precoding action a_t through a single forward pass. The world model predictors and critic are not used online.

C. Complexity Analysis

We analyze the computational complexity of GWM-RA during offline training and online inference.

Offline training requires substantial computation, dominated by training the graph-based world model. For each training step, the forward pass of the L -layer GNN encoder has complexity approximately $\mathcal{O}(L|\mathcal{E}|d_{\text{feat}}^2)$, where $|\mathcal{E}|$ is the number of edges in $\mathcal{G}_t = (\mathcal{V}, \mathcal{E})$ and d_{feat} is the maximum hidden-feature dimension. The subsequent actor-critic updates operate on the latent representation and are dominated by the forward and backward passes through the MLP-based actor and critic. This computation is performed during server-side training and can be parallelized.

During online inference, GWM-RA generates the digital-precoding action for a channel snapshot using one forward pass through the trained GNN encoder and actor, followed by the action mapping in (23). The online complexity is therefore dominated by the GNN encoder, i.e., $\mathcal{O}(L|\mathcal{E}|d_{\text{feat}}^2)$, together with the actor forward pass and the normalization and bounding operations in (23). This procedure is non-iterative. In contrast, conventional optimization methods such as WMMSE require multiple iterations to converge, and each iteration involves matrix operations with complexity up to $\mathcal{O}(|\mathcal{K}_s|^3 N_{\text{RF}}^3)$ under the considered implementation. The resulting computational delay increases with the number of users, supporting the use of GWM-RA for low-latency digital precoding in dynamic LEO satellite networks.

IV. SIMULATION AND RESULTS ANALYSIS

This section presents numerical results to evaluate the performance of the proposed GWM-RA method, with system-wide EE as the primary metric. We begin by detailing the simulation setup and the baseline algorithms used for comparison. Subsequently, a comprehensive performance evaluation is provided, followed by a validation of the key architectural advantages concerning sample efficiency and computational cost. Finally, an ablation study is conducted to verify the contributions of the individual components of the GWM-RA.

A. Simulation Setup and Baseline Algorithms

The simulation environment implements the multi-satellite LEO network, channel models, and power-consumption models described in Section II. The key simulation parameters, chosen based on 3GPP NTN standards and representative values from recent literature, are summarized in Table II. Unless otherwise stated, all results are averaged over 20 independent random seeds, and shaded bands indicate 95% confidence intervals (CIs).

The performance of the proposed GWM-RA is benchmarked against a set of baseline algorithms, which are organized into two distinct categories for a systematic comparison: iterative optimization methods and model-free reinforcement learning approaches.

1) *Iterative Optimization Baselines*: This category represents conventional, non-learning approaches and serves to benchmark our method against established optimization techniques.

- *WMMSE-based EE (WMMSE-EE)*: An iterative algorithm adapted from the weighted minimum mean-square error (WMMSE) method, combined with the Dinkelbach procedure to handle the fractional EE objective. It requires full instantaneous CSI and multiple iterations to converge. This baseline serves as a strong high-complexity iterative reference for conventional optimization-based methods.
- *Zero-forcing-based EE (ZF-EE)*: A low-complexity linear precoding scheme in which beamforming directions are determined by the zero-forcing (ZF) criterion, followed by power allocation to maximize EE. It represents a practical heuristic for interference management.

2) *Model-Free Reinforcement Learning Baselines*: This category is designed to validate the core contributions of our model-based and graph-aware approach through controlled ablation. The two model-free baselines are standard actor-critic references under the same LEO-HBF state/action interface: GNN-MFRL shares the topology-aware encoder with GWM-RA, whereas MLP-MFRL removes this graph-structured inductive bias. This setup separates the gain from topology-aware encoding from the gain brought by surrogate-assisted policy improvement, while enhanced model-free designs are discussed together with the sample-efficiency results in Section IV-C.

- *GNN-based model-free RL (GNN-MFRL)*: An actor-critic agent that uses a GNN encoder identical to that of GWM-RA but learns end-to-end through direct environment interaction. By comparing GWM-RA with

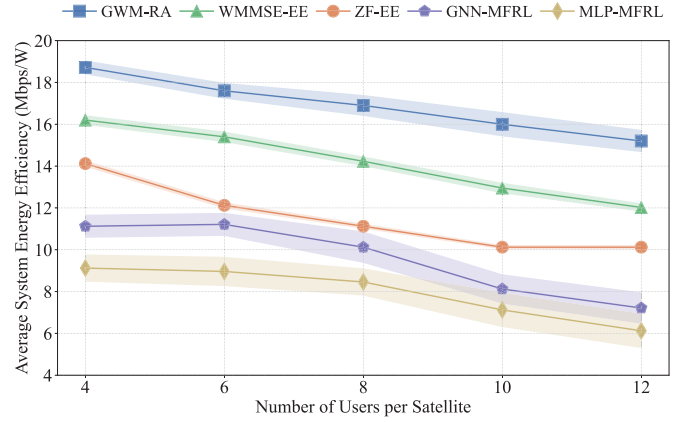


Fig. 3. Average system energy efficiency vs. the number of users per satellite. Lines show mean values; shaded bands indicate 95% confidence intervals over 20 seeds.

GNN-MFRL, we isolate and quantify the performance and sample-efficiency gains attributable to the proposed performance-surrogate component under the adopted training protocol.

- *MLP-based model-free RL (MLP-MFRL)*: A standard model-free RL agent whose policy is encoded by an MLP that takes a flattened channel vector as input, thus ignoring the network topology. Comparing GNN-MFRL against MLP-MFRL allows us to quantify the benefits of employing a GNN for topology-aware state representation.

B. Performance Evaluation and Analysis

This section evaluates the performance of the proposed GWM-RA against the established baselines. We first compare the converged energy efficiency under various network conditions and then quantify the sample-efficiency and computational advantages of the proposed method in Section IV-C.

The scalability of the algorithms with respect to network load and interference complexity is evaluated in Fig. 3. GWM-RA achieves the highest EE among the considered baselines, and its margin over GNN-MFRL widens as the number of users increases. This trend is consistent with the graph-conditioned world model capturing topology-action-performance coupling in dense interference regimes. The persistent gap between GNN-MFRL and MLP-MFRL also supports the benefit of topology-aware encoding.

Fig. 4 evaluates performance under varying maximum satellite power budgets. GWM-RA maintains the highest EE among the considered methods throughout. For all methods, EE first increases with $P_s^{\max}$ in the power-limited regime and then saturates as the system becomes interference-limited. The larger gap in the high-power regime suggests a more favorable rate-power trade-off for GWM-RA in the evaluated setting.

C. Sample Efficiency and Computational Scalability

This section quantifies the two primary design advantages of the proposed GWM-RA: the sample efficiency enabled by

TABLE III
EMPIRICAL QUANTIFICATION OF TRAINING AND INFERENCE COSTS

Metric	GWM-RA (Proposed)	GNN-MFRL	MLP-MFRL	WMMSE-EE
Env. Interactions to Threshold (12.5 Mbps/W)	0.8×10^6	2.2×10^6	N/A	N/A
Env. Interactions to Convergence	1.9×10^6	3.3×10^6	4.5×10^6	N/A
Total Gradient Updates (World Model)	$\sim 1.5 \times 10^5$	N/A	N/A	N/A
Total Gradient Updates (Actor/Critic)	$\sim 1.5 \times 10^6$	$\sim 2.5 \times 10^5$	$\sim 3.5 \times 10^5$	N/A
Wall-clock training time	2.8 h	1.7 h	1.2 h	N/A
Online Inference Time (per slot, user load = 8)	1.2 ms	1.2 ms	0.4 ms	~ 1350 ms

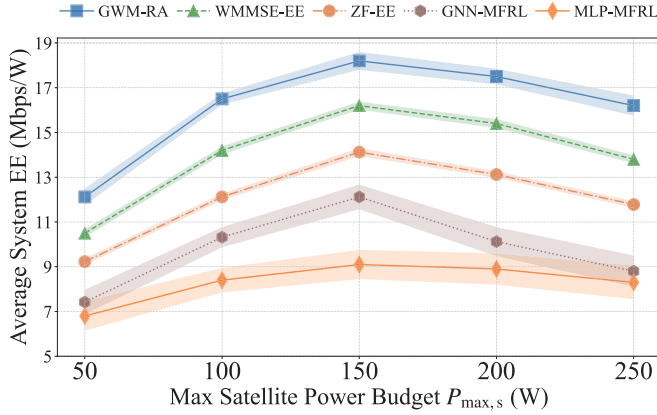


Fig. 4. Average system energy efficiency vs. the maximum satellite power budget. Lines show mean values; shaded bands indicate 95% confidence intervals over 20 seeds.

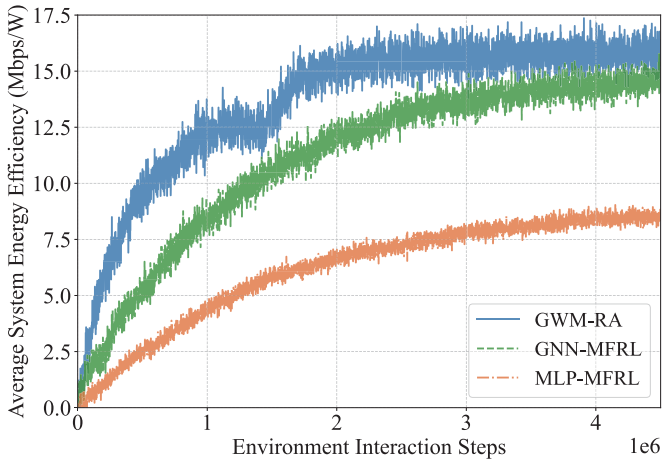


Fig. 5. Sample efficiency comparison of the proposed GWM-RA, the topology-aware GNN-MFRL baseline, and the topology-agnostic MLP-MFRL baseline. Lines show mean values; shaded bands indicate 95% confidence intervals over 20 seeds.

its model-based nature, and the low computational complexity afforded by its non-iterative inference process.

A primary advantage of the model-based paradigm is improved sample efficiency. Fig. 5 compares GWM-RA with both model-free baselines, where including MLP-MFRL disentangles gains from topology-aware encoding and from the performance-surrogate component. GNN-MFRL requires about 2.2×10^6 interactions to reach 12.5 Mbps/W, whereas GWM-RA reaches the same level with only 0.8×10^6 ,

i.e., over 60% fewer interactions; MLP-MFRL does not reach this threshold. GWM-RA also converges faster, at about 1.9×10^6 interactions versus 3.3×10^6 for GNN-MFRL and 4.5×10^6 for MLP-MFRL. Within this controlled comparison, GNN encoding raises the achievable performance level, while the learned performance surrogate improves sample efficiency by enabling one-step surrogate-assisted policy updates and reducing the number of true-environment reward evaluations under the considered training protocol.

The comparison in Fig. 5 should also be interpreted in light of the model-free baselines evaluated in this paper. Recent model-free RL methods can improve exploration and convergence through intrinsic rewards, attention modules, or refined training strategies [24]. Such mechanisms may reduce the empirical interaction gap relative to standard GNN-MFRL when carefully tailored to the target system. They are also complementary to the proposed framework: curiosity-driven exploration or attention-based policies could be incorporated into the actor or graph encoder, whereas GWM-RA provides a supervised graph-conditioned rate/power surrogate for one-step policy improvement. The evidence in Fig. 5 and Table III therefore supports the sample-efficiency benefit of the proposed performance-surrogate design for the considered baselines and training setting. A systematic comparison with enhanced model-free agents, as well as their integration with GWM-RA, is left for future work.

Table III reports empirical training and inference costs measured on a server with dual NVIDIA RTX 4090 GPUs and 64-thread Intel Xeon CPUs. GWM-RA converges with 1.9×10^6 interactions but requires 2.8 training hours, versus 1.7 hours for GNN-MFRL, due to additional world-model training ($\sim 1.5 \times 10^5$ updates) and imagination-based actor/critic optimization ($\sim 1.5 \times 10^6$ updates). In deployment, its online inference remains lightweight at 1.2 ms per slot: it matches the GNN-MFRL baseline, is slower than the simpler MLP-MFRL baseline (0.4 ms), and remains far below the ~ 1350 ms delay of WMMSE-EE. These wall-clock measurements should be interpreted as offline server-side reference costs under the adopted training protocol rather than as direct estimates of on-board satellite training or inference runtime.

Beyond sample efficiency, real-time operation also requires low and scalable inference complexity. Fig. 6 shows that the inference time of WMMSE-EE increases rapidly as the per-satellite user load in the evaluated local scenario grows. Here, the horizontal axis indicates the number of users present in the evaluated local scenario per satellite, whereas the simultaneously served short-slot streams in deployment remain limited

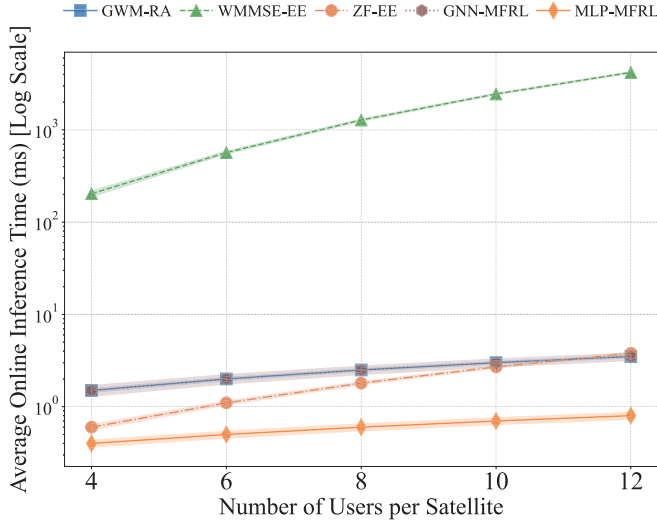


Fig. 6. Average online inference time vs. the number of users per satellite. Lines show mean values; shaded bands indicate 95% confidence intervals over 20 seeds.

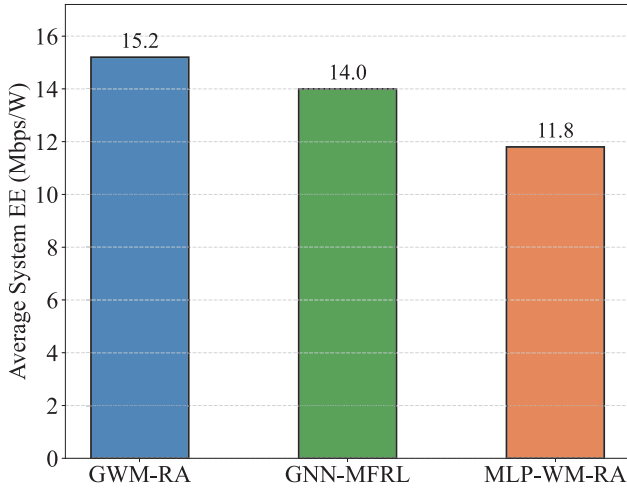


Fig. 7. Ablation study on the importance of the encoder.

by N_{RF} . The measured inference time of GWM-RA increases approximately linearly over the evaluated range, consistent with the analysis in Section III-C.

D. Ablation Study

Two ablation studies were conducted. Unless otherwise stated, the results are reported under the default training and evaluation setting adopted for the ablation study.

First, the contributions of the model-based paradigm and the topology-aware GNN encoder are evaluated through the ablation study presented in Fig. 7. The effect of the GNN encoder is examined by comparing GWM-RA with its MLP-WM-RA variant, in which the GNN encoder is replaced by a standard MLP. This replacement reduces EE from 15.2 Mbps/W to 11.8 Mbps/W, corresponding to a 22.4% decrease relative to GWM-RA. The observed performance degradation indicates that explicitly representing dynamic nodes, edges, and their structural dependencies is beneficial in the evaluated scenarios.

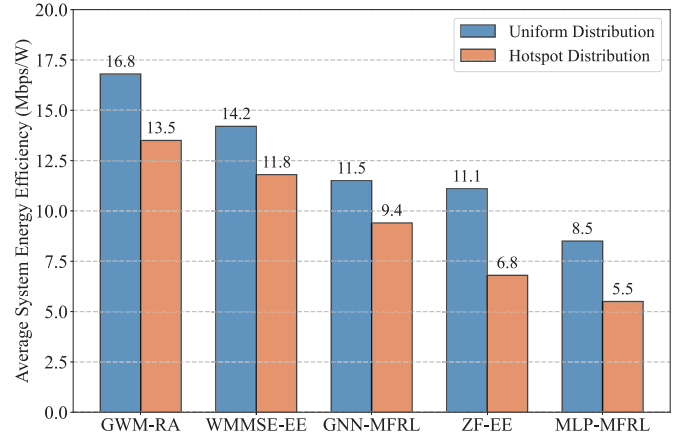


Fig. 8. Cross-scenario performance validation: Average system energy efficiency under uniform user distribution versus severe non-uniform hotspot distribution.

The message-passing mechanism provides an inductive bias for characterizing non-local interference and constructing the world model.

The effect of the model-based paradigm is further evaluated by comparing GWM-RA with the model-free GNN-MFRL baseline. Both methods use the same GNN encoder, while GWM-RA achieves 15.2 Mbps/W, an 8.6% improvement over the 14.0 Mbps/W achieved by GNN-MFRL. Under this controlled comparison, the difference is associated with the learned rate/power surrogate, which provides additional one-step reward and constraint signals for actor/critic updates under the same GNN encoder.

The second ablation evaluates the constraint-aware penalty terms in the imagined reward function. The GWM-RA (Pure EE) variant was trained using an imagined reward that maximized only the raw EE objective ($\hat{R}_t/\hat{P}_t$), without penalty terms. Although GWM-RA (Pure EE) achieves higher raw EE, its QoS and power-constraint violation rates increase to 27.3% and 18.9%, respectively. In contrast, the penalty-augmented GWM-RA limits these rates to 0.8% and 0.5%. These results show that the penalty terms guide the learned policy toward empirically near-feasible solutions of (P1). Although they do not provide a formal hard-feasibility guarantee, they substantially reduce the observed QoS and power violations.

E. Cross-Scenario Validation: Non-Uniform User Distributions

To examine robustness beyond the default uniform setting, we consider a severe hotspot scenario in which 70% of scheduled users are concentrated within 30% of the coverage area, intensifying both inter-beam and inter-satellite interference.

For $|\mathcal{K}_s| = 8$, all methods degrade, as shown in Fig. 8 and Table V. ZF-EE suffers the largest drop (38.7%), and MLP-MFRL falls to 5.5 Mbps/W.

GWM-RA still achieves the highest hotspot EE, reaching 13.5 Mbps/W. The GNN-MFRL baseline attains 9.4 Mbps/W, supporting the benefit of topology-aware encoding, while the remaining gap to GWM-RA is consistent with the contribution of the performance-surrogate component in the evaluated

TABLE IV
IMPACT OF CONSTRAINT-AWARE REWARD

Algorithm	Avg. EE (Mbps/W)	Violation Rate (%)	
		QoS	Power
GWM-RA (Pen.)	15.2	0.8	0.5
GWM-RA (Pure EE)	16.5	27.3	18.9

TABLE V
PERFORMANCE DEGRADATION ACROSS DISTRIBUTIONS

Algorithm	Uniform EE (Mbps/W)	Hotspot EE (Mbps/W)	EE Degradation (%)
GWM-RA (Proposed)	16.8	13.5	19.6%
WMMSE-EE (Iterative)	14.2	11.8	16.9%
GNN-MFRL	11.5	9.4	18.3%
ZF-EE (Zero-Forcing)	11.1	6.8	38.7%
MLP-MFRL (Topology-Agnostic)	8.5	5.5	35.3%

hotspot scenario. The uniform-distribution values in Fig. 8 and Table V correspond to the dedicated cross-scenario evaluation setting with $|\mathcal{K}_s| = 8$, and are therefore not expected to match the ablation-study operating point exactly.

V. CONCLUSION AND FUTURE WORK

This paper studied energy-efficient HBF in dynamic LEO satellite networks and proposed GWM-RA, which uses a GNN-based, action-conditioned surrogate of user-wise rates and total power for surrogate-assisted, non-iterative short-slot digital precoding under fixed long-timescale analog precoding and scheduling. Simulations showed higher EE than the considered benchmarks together with reduced environment-interaction demand during policy adaptation under the adopted training protocol.

Future work includes extending the surrogate with sequence modeling for non-i.i.d. channel evolution, developing compression or quantization methods for efficient on-board deployment, and studying how enhanced model-free exploration mechanisms can be integrated with the proposed performance-surrogate framework.

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